

Questions with Answer Keys

MathonGo

Q1 (20 July 2021 Shift 1)

If $\vec{A}$ and $\vec{B}$ are two vectors satisfying the relation $\vec{A} \cdot \vec{B} = |\vec{A} \times \vec{B}|$. Then the value of $|\vec{A} - \vec{B}|$ will be

- (1) $\sqrt{A^2 + B^2}$
- (2) $\sqrt{A^2 + B^2 + \sqrt{2}AB}$
- (3) $\sqrt{A^2 + B^2 + 2AB}$
- (4) $\sqrt{A^2 + B^2 - \sqrt{2}AB}$

Q2 (20 July 2021 Shift 2)

Two vectors $\vec{P}$ and $\vec{Q}$ have equal magnitudes. If the magnitude of $\vec{P} + \vec{Q}$ is n times the magnitude

of $\vec{P} - \vec{Q}$, then angle between $\vec{P}$ and $\vec{Q}$ is :

- (1) $\sin^{-1}\left(\frac{n-1}{n+1}\right)$
- (2) $\cos^{-1}\left(\frac{n-1}{n+1}\right)$
- (3) $\sin^{-1}\left(\frac{n^2-1}{n^2+1}\right)$
- (4) $\cos^{-1}\left(\frac{n^2-1}{n^2+1}\right)$

Q3 (22 July 2021 Shift 1)

What will be the projection of vector $\vec{A} = \hat{i} + \hat{j} + \hat{k}$ on vector $\vec{B} = \hat{i} + \hat{j}$?

- (1) $\sqrt{2}(\hat{i} + \hat{j} + \hat{k})$
- (2) $2(\hat{i} + \hat{j} + \hat{k})$
- (3) $\sqrt{2}(\hat{i} + \hat{j})$
- (4) $(\hat{i} + \hat{j})$

Q4 (22 July 2021 Shift 1)

Three particles P, Q and R are moving along the vectors $\vec{A} = \hat{i} + \hat{j}$, $\vec{B} = \hat{j} + \hat{k}$ and $\vec{C} = -\hat{i} + \hat{j}$ respectively.

They strike on a point and start to

move in different directions. Now particle P is moving normal to the plane which contains vector $\vec{A}$ and $\vec{B}$. Similarly

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particle Q is moving normal to the plane which contains vector $\vec{A}$ and $\vec{C}$. The angle between the direction of motion of P and Q is $\cos^{-1}\left(\frac{1}{\sqrt{x}}\right)$. Then the value of x is _____.

Q5 (22 July 2021 Shift 1)

Three students S_1 , S_2 and S_3 perform an experiment for determining the acceleration due to gravity (g) using a simple pendulum. They use different lengths of pendulum and record time for different number of oscillations. The observations are as shown in the table.

Student No.	Length of pendulum (cm)	No. of oscillations (n)	Total time for n oscillations	Time period (s)
1.	64.0	8	128.0	16.0
2.	64.0	4	64.0	16.0
3.	20.0	4	36.0	9.0

(Least count of length = 0.1 m least count for time = 0.1 s) If E_1 , E_2 and E_3 are the percentage errors in ' g ' for students 1, 2 and 3 respectively, then the minimum percentage error is obtained by student no. _____.

Q6 (25 July 2021 Shift 1)

Match List I with List II.

List I		List II	
(a)	$\vec{C} - \vec{A} - \vec{B} = 0$	(i)	
(b)	$\vec{A} - \vec{C} - \vec{B} = 0$	(ii)	
(c)	$\vec{B} - \vec{A} - \vec{C} = 0$	(iii)	
(d)	$\vec{A} + \vec{B} = -\vec{C}$	(iv)	

Choose the correct answer from the options given below :

(1) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (i), (c) $\rightarrow$ (iii), (d) $\rightarrow$ (ii)

(2) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (i), (d) $\rightarrow$ (ii)

(3) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (i)

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(4) (a) $\rightarrow$ (i), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (iii)

Q7 (25 July 2021 Shift 2)

Two vectors $\vec{X}$ and $\vec{Y}$ have equal magnitude. The magnitude of $(\vec{X} - \vec{Y})$ is n times the magnitude of $(\vec{X} + \vec{Y})$. The angle between $\vec{X}$ and $\vec{Y}$ is :

(1) $\cos^{-1}\left(\frac{-n^2-1}{n^2-1}\right)$

(2) $\cos^{-1}\left(\frac{n^2-1}{-n^2-1}\right)$

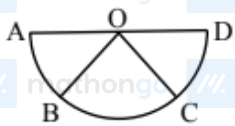
(3) $\cos^{-1}\left(\frac{n^2+1}{-n^2-1}\right)$

(4) $\cos^{-1}\left(\frac{n^2+1}{n^2-1}\right)$

Q8 (27 July 2021 Shift 1)

Assertion A : If A, B, C, D are four points on a semi-circular arc with centre at 'O' such that $|\vec{AB}| = |\vec{BC}| = |\vec{CD}|$, then

$$\vec{AB} + \vec{AC} + \vec{AD} = 4\vec{AO} + \vec{OB} + \vec{OC}$$

Reason R : Polygon law of vector addition yields $\vec{AB} + \vec{BC} + \vec{CD} + \vec{AD} = 2\vec{AO}$ 

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) **A** is correct but **R** is not correct.(2) **A** is not correct but **R** is correct.(3) Both **A** and **R** are correct and **R** is the correct explanation of **A**.(4) Both **A** and **R** are correct but **R** is not the correct explanation of **A**.

Q9 (27 July 2021 Shift 2)

A physical quantity 'y' is represented by the formula $y = m^2 r^{-4} g^x l^{-\frac{3}{2}}$

If the percentage errors found in y, m, r, l and g are 18, 1, 0.5, 4 and p respectively, then find the value of x and p.

(1) 5 and ± 2 (2) 4 and ± 3 (3) $\frac{16}{3}$ and $\pm \frac{3}{2}$

Questions with Answer Keys

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Answer Key

Q5 (1) **Q6 (2)** **Q7 (2)** **Q8 (4)**

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Hints and Solutions

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Q1

$$\vec{A} \cdot \vec{B} = |\vec{A} \times \vec{B}|$$

$$AB \cos \theta = AB \sin \theta \Rightarrow \theta = 45^\circ$$

$$|\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 - 2AB \cos 45^\circ}$$

$$= \sqrt{A^2 + B^2 - \sqrt{2}AB}$$

Hence option (4).

Q2

$$|\vec{P}| = |\vec{Q}| = x \dots (i)$$

$$|\vec{P} + \vec{Q}| = n|\vec{P} - \vec{Q}|$$

$$P^2 + Q^2 + 2PQ \cos \theta = n^2 (P^2 + Q^2 - 2PQ \cos \theta)$$

Using (i) in above equation

$$\cos \theta = \frac{n^2 - 1}{1 + n^2}$$

$$\theta = \cos^{-1} \left(\frac{n^2 - 1}{n^2 + 1} \right)$$

Q3

$$(A \cos \theta) \hat{B} = A \left(\frac{\vec{A} \cdot \vec{B}}{AB} \right) \hat{B} = \frac{\vec{A} \cdot \vec{B}}{B} \hat{B}$$

$$= \frac{2}{\sqrt{2}} \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right) = \hat{i} + \hat{j}$$

Q4

$$\text{Direction of } P\hat{v}_1 = \pm \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|} = \pm \frac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$$

$$\text{Direction of } Q\hat{v}_2 = \pm \frac{\vec{A} \times \vec{C}}{|\vec{A} \times \vec{C}|} = \pm \frac{2\hat{k}}{2} = \pm \hat{k}$$

Angle between $\hat{v}_1$ and $\hat{v}_2$

$$\frac{\hat{v}_1 \cdot \hat{v}_2}{|\hat{v}_1| |\hat{v}_2|} = \frac{\pm 1/\sqrt{3}}{(1)(1)} = \pm \frac{1}{\sqrt{3}}$$

$$\Rightarrow x = 3$$

Q5

Hints and Solutions

MathonGo

$$T = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow g = \frac{4\pi^2\ell}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + \frac{2\Delta T}{T}$$

$$\Delta T = \frac{\text{least count of time } (\Delta T_0)}{\text{number of oscillations}(n)}$$

$$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + \frac{2\Delta T_0}{nT}$$

As $\Delta \ell$ and ΔT_0 are same for all observations so

$$\frac{\Delta g}{g} \text{ is minimum for highest value of } \ell, n \text{ and } T$$

$\Rightarrow$ Minimum percentage error in g is for student number-1

Q6

$$(a) \vec{C} = \vec{A} + \vec{B}$$

Option (iv)

$$(b) \vec{A} = \vec{B} + \vec{C} = \vec{C} + \vec{B}$$

Option (iii)

$$(c) \vec{B} = \vec{A} + \vec{C}$$

Option (i)

$$(d) \vec{A} + \vec{B} + \vec{C} = 0$$

Option (ii)

Q7

Given $X = Y$

$$\sqrt{X^2 + Y^2 - 2 \times Y \cos \theta}$$

$$= n\sqrt{X^2 + Y^2 + 2 \times Y \cos \theta}$$

$$\text{Square both sides } 2X^2(1 - \cos \theta) = n^2 \cdot 2X^2(1 + \cos \theta)$$

$$1 - \cos \theta = n^2 + n^2 \cos \theta$$

$$\cos \theta = \frac{1-n^2}{1+n^2}$$

$$\theta = \cos^{-1} \left[\frac{n^2-1}{-n^2-1} \right]$$

Q8

Polygon law is applicable in both but the equation given in the reason is not useful in explaining the assertion.

Hints and Solutions

MathonGo

Q9

$$\frac{\Delta y}{y} = \frac{2\Delta m}{m} + \frac{4\Delta r}{r} + \frac{x\Delta g}{g} + \frac{3}{2} \frac{\Delta \ell}{\ell}$$

$$18 = 2(1) + 4(0.5) + xp + \frac{3}{2}(4)$$

$$8 = xp$$

By checking from options, $x = \frac{16}{3}$, $p = \pm \frac{3}{2}$